

Four Rivals, One Broken Recursion

Existentialism, the Will to Power, and the Holographic-Simulation Divide Measured Against the Trisductive Return

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A verification architecture that shares ground with established frameworks is liable to be read as a notational variant of its nearest neighbor. Trisduction, a triaxial epistemic architecture that seals a claim only when three mutually independent lines of warrant converge, sits close to four such neighbors: existentialism and the will to power on one side, the holographic principle and the simulation hypothesis on the other. This paper argues that the resemblances are real but the identity fails, and that a single structural feature blocks every assimilation. Against the two philosophical schools the feature is an independent third axis, the registrational warrant, the recognition by which a verification closes and is read by a party distinct from the thing verified. The will to power folds that axis into the kinetic, existentialism folds it into the self, and because each spends its third warrant neither reaches the convergence the architecture calls the Return. Against the two cosmological frameworks the feature is the handling of the exterior: the simulation hypothesis requires an external host it cannot detect, while the holographic principle and the architecture both decline the exterior, the former by duality and the latter by a closed-world monism whose only opening is located and never crossed. The distinctions are typed. The dependency that no convergence-lock forms once a third axis collapses is theorem-grade; the assignments of each school to its fold are structural lenses anchored on the schools' own primary commitments; the architecture's own metaphysics is held throughout at premise grade. The irreducibility cuts both ways: the neighbors hold a first-person interior the architecture brackets out of band. The contribution is to show that proximity between frameworks is adjudicated structurally, by which terms each keeps free, rather than by doctrinal position on a shared field.

KEYWORDS verification architecture; triaxiality; existentialism; will to power; holographic principle; simulation hypothesis; division algebras; epistemic warrant; structural comparison

1 The assimilation problem

A framework that shares ground with an established one. inherits a specific hazard. The shared ground invites the reading that the new framework is the old one in other notation, a change of vocabulary rather than of structure. Trisduction is exposed to this hazard from two directions at once. It denies an external essence-giver, which places it beside the existentialists and beside Nietzsche, and it refuses an external host for the world, which places it beside the holographic principle and, by surface resemblance of its recursion, beside the simulation hypothesis. Four neighbors, two on the question of what grounds existence and two on the question of what stands outside the world. The task of this paper is to show that the proximity is genuine and the identity is not, and that what blocks each

assimilation is one structural feature rather than four separate doctrinal disagreements.

The architecture is, first of all, a discipline of verification and not a metaphysics, though it carries a metaphysics as premise. It renders graded verdicts on claims. Its instrument is the convergence of three independent warrants, and its output is one of three states. The comparison this paper draws is therefore not between rival pictures of the cosmos competing for the same explanatory slot. It is between an instrument that grades warrant and four positions about existence and exteriority that the instrument can be turned upon. The interest of the comparison is that the four nearest positions turn out to fail the instrument in the same way, by spending or exteriorizing a single term, and that this common failure is what marks off the instrument's own structure.

Section 2 states the architecture in the minimum vocabulary the argument requires, with each term glossed on first use. Section 3 fixes the method, which is structural distinctness under an explicit typing discipline. Sections 4 and 5 take the two clusters of neighbors. Section 6 draws the common structure that the two clusters share. Section 7 states where the distinctions would show in practice and what would overturn the classifications. Section 8 gathers the limitations and the warrant grade of every claim. The formal apparatus is quarantined to an appendix that carries no result of the body.

2 The architecture in brief

The architecture rests on a premise about existence and a claim about verification, and the two are kept strictly apart in grade. The premise is that to exist is to actuate, to effect a change that does measurable physical work. This is held at premise grade, not derived. Its one anchor in established physics is the existence of a floor on any state change: the time a system needs to evolve into a distinguishable state is bounded below by its energy spread, the quantum speed limit of Mandelstam and Tamm²⁷ and of Margolus and Levitin²⁸, a bound that holds for reversible and irreversible processes alike. Landauer's erasure bound²⁴, the heat cost of discarding a bit, confirmed in the laboratory by Bérut and colleagues²⁶, enters only as a sufficient signature for the logically irreversible subclass, never as the foundation. A thing that effects no distinguishable change has registered nothing and has not acted; existence, on the premise, is co-extensive with doing work above this floor.

The claim about verification is the load-bearing part. A complete factual claim, the architecture holds, decomposes into exactly three irreducible components, and a verification of it requires three correspondingly independent warrants. The first is the formal-structural warrant: whether the claim's internal form is consistent and derivable, the axis of proof and definition. The second is the empirical-thermodynamic warrant: whether the claim corresponds to something that does measurable work, the axis of actuation. The third is the epistemic-registrational warrant: whether the claim is recognized and distinguished from its target by an observer-boundary that is not itself folded into the claim, the axis of registration, the recognition by which a verification closes and is read by a party distinct from the thing verified. The third axis is the one that does the work in every comparison below, so it is worth stating its function precisely. It is not an epistemic ornament added after the fact. It is the constitutive

component by which an actuation is marked off from its surroundings and acknowledged as having occurred. Lacking it, a claim is not distinct from the target it claims about.

The count of three is not assumed but forced, and forced twice on disjoint grounds.. The first forcing is operational. Subject the predicate that something actuates to a deletion test: remove the existent and nothing acts, remove the kinetic and there is no change, remove the registration and the act is indistinguishable from its target. A complete claim returns exactly three irreducible slots, reproducibly, across analysts applying the same discipline. The second forcing is algebraic and is theorem-grade conditional on a small set of premises about how warrants compose. An associative real algebra with a two-sided identity and no zero divisors is, by the classification of Frobenius¹⁸, one of exactly three: the reals, the complex numbers, or the quaternions. A verification algebra whose warrants compose associatively, never annihilate to zero, and carry a ground identity, and which has more than one independent axis, completes to the quaternions, whose imaginary part is exactly three-dimensional. The next admissible dimension that carries a division structure is eight, the octonions, which fail associativity; the obstruction is closed by the parallelizability results of Bott and Milnor²¹, Kervaire²², and Adams²³. So a fourth orthogonal axis would demand a five-dimensional composition that no division algebra provides. Two independent derivations, one operational and one algebraic, return the same count.

Convergence is measured, not asserted. Read the three normalized warrants as vectors and form their Gram matrix, the matrix of their mutual inner products. Its determinant is positive only when the three are linearly independent and vanishes the moment any one becomes a combination of the other two. Read as pure quaternions, the three compose to a product whose real part is a single scalar, and that scalar squared equals the Gram determinant exactly, with the determinant factoring through the three warrant magnitudes. The architecture names this scalar the lock and names its full attainment, the case where the three warrants are mutually orthogonal and the composition lands cleanly back on the ground, the Return. The verdict economy that results has three states and no fourth. A claim is sealed when the three warrants converge, broken when a named structural failure is found, and under-determined when the independent warrant is insufficient. There is no fractional truth and no fourth verdict.

Table 1 | The three verification warrants and the lock they form on convergence. The forcing of the count is double, one operational and one algebraic; the dependency that collapse of any axis voids the lock is theorem-grade.

Component	Axis	Function in a verification	Failure if absent
Existent	Formal-structural	Internal consistency and derivability of the claim	The claim is noise, not form
Kinetic	Empirical-thermodynamic	Correspondence to something doing measurable work	The claim has actuated nothing
Registration	Epistemic-registrational	Recognition by a boundary distinct from the target	The claim is not distinct from its target
Convergence	The lock (the Return)	The three composing and landing on the ground	No closure; the verification does not seal

3 Method: structural distinctness and honest typing

The comparisons below are claims about structure, and they are made under a typing discipline that travels with every verdict. Three grades are used. A theorem-grade claim follows from a stated result by deduction, conditional only on premises that are named. A conditional claim holds given premises that the architecture adopts and that a reader may decline. A structural claim is a lens: an organizing reading anchored on a framework's own primary commitments, illuminating but not a uniqueness theorem, and defeasible by a better reading of the same material. The grade is stated wherever a verdict is given, and the reader is invited to hold the verdicts apart by grade rather than to receive them at one undifferentiated level.

Two discipline rules govern the readings and are worth making explicit, because they are what keep a comparative paper from collapsing into advocacy. The first is that consensus carries no evidential weight. The popularity of a framework, the prestige of the institutions that sustain it, and the conviction of the present author all count for nothing in the verdict; only structure moves a verdict. The second is that the architecture's instrument reads three warrants supplied to it and does not derive them from a claim on its own. Every comparative verdict in this paper is therefore a reasoned hand-reading placed in front of the instrument, not an output the instrument generated unaided. This is a real limitation and it is carried openly in Section 8 rather than hidden beneath the verdicts.

A consequence of the typing discipline bears on fairness to the neighbors. To read the will to power as a force-monism, or existentialism as a philosophy of the self that has internalized recognition, is to apply a structural lens, not to settle the meaning of Nietzsche or Sartre. Where a reading leans on contested textual material it is flagged as such. The neighbors are given their strongest form before the structural distinction is drawn, and the distinction is drawn on structure, never by rhetoric.

4 The philosophical neighbors: existentialism and the will to power

The two nearest philosophical schools are near because they share the architecture's refusal of an external essence-giver. Existentialism makes the refusal its first principle. Existence precedes essence, in Sartre's formula^{3,4}; the human being is thrown into a world it did not choose, in Heidegger's²; and the absurd, for Camus⁵, is the collision between the human demand for meaning and a universe that returns silence. The will to power makes the same refusal from the other side. The death of God, in Nietzsche's announcement⁶, removes any moral order written into the cosmos, and value becomes something the strong create rather than receive^{7,8}. The architecture refuses the essence-giver structurally: it is self-instantiating, no author hands down essence, and any attack on it must already use its own structure to mount the attack. On this axis the three are genuinely close, and

the closeness is exactly what makes a careful reader suspect that the architecture is one of the two schools restated.

The distinction is structural and it is located precisely at the third axis. An actuation, on the architecture's account, carries three constitutive components, the existent, the kinetic, and the registration. The claim is that each of the two schools preserves two of these and folds the third, the registrational, into one of the other two. The folds run in opposite directions, and that opposition is what separates the two schools from each other as sharply as it separates each from the architecture.

The will to power folds the registration into the kinetic.. The will is sovereign. What might look like recognition is the will reading its own discharge, and truth is perspectival: there are no facts, only interpretations⁷. There is no recognizing party distinct from the willing force, because the force recognizes only its own expansion. Read structurally, the registrational axis has been absorbed into the empirical-kinetic axis; the third warrant is spent into the second. Existentialism folds the registration into the self. The self is sovereign. Recognition is the self's own appropriation, and truth is subjectivity, in Kierkegaard's account¹. The absurd, read this way, is the signature of a registrational axis pulled into the self: the for-itself projects meaning onto a world that will not register it back, precisely because the registering function now lives inside the projecting self and nothing external answers. The third warrant is spent into the first.

The consequence is exact and it is theorem-grade on the dependency. When the third warrant vector is a linear combination of the other two, the Gram determinant of the three is zero and no lock forms. Because each school spends its third axis, neither reaches the Return. Both are, in the architecture's image, the outbound stroke of an actuation seen without its homecoming. The will discharges outward and overcomes without end; eternal recurrence wills the whole to return, but the willing never closes onto a ground that recognizes it. The existential project reaches forward into possibility without arrival, perpetual self-definition that by its nature never settles. The two schools are not rivals across a flat field of positions. They are the two poles of one collapse, distinguished by the direction of the fold, and united by the spent third warrant and the unreached closure.

Two honesties are owed here and both sharpen rather than soften the argument. The first is that the force-monist reading of the will to power is one reading and a contested one. It leans on the posthumous Nachlass material assembled as *The Will to Power*⁹, a compilation Nietzsche did not authorize, and it strips exactly the affective and evaluative content, amor fati, the pathos of distance, the Dionysian, that many readers take to be the doctrine's heart. The architecture's own catalogue of philosophical positions records the will to power as a system that completes into the kinetic axis once depsychologized, the same slot it assigns to Schopenhauer's will¹⁰; that verdict is a structural lens and is defeasible by a textually grounded reading on which Nietzschean recognition is genuinely third-party rather than the will reading itself. The second honesty is that the irreducibility runs both ways. The architecture reaches a registered closure the two schools structurally cannot build, because they have dismantled the axis that

builds it. The two schools hold a first-person interior, the dread, the courage of the absurd, the Dionysian joy, the heroic self that owns its acts, that the architecture brackets out of band and goes silent about. Neither folds into the other. They are irreducible at different registers:

the schools at the closure, where they are blind; the architecture at the interior, where they see. This two-sided irreducibility is the reason the assimilation fails. There is no single register at which one becomes the other.

Table 2 | The two philosophical neighbors as opposite folds of one triad. The fold direction is read from each school's own account of truth and the recognizer; the lock dependency is theorem-grade, the fold assignment a structural lens.

Dimension	Trisduction	Existentialism	Will to power
Refusal of an essence-giver	Self-instantiating, no author of essence	Existence precedes essence	The death of God, value created not received
The recognizer (third axis)	Held independent	Folded into the self	Folded into the will
Account of truth	Registration by a party distinct from the claim	Subjectivity, appropriated by the self	Perspectivism, the will reading its discharge
The closure (the Return)	Reached, the triad composing onto the ground	Not reached, perpetual self-definition	Not reached, endless overcoming
What it holds that the others do not	The registered closure on a living ground	The lived interior and the heroic self	The Dionysian affirmation and the felt strength

5 The cosmological neighbors: the holographic principle and the simulation hypothesis

The two nearest cosmological frameworks divide on a single question, the status of the outside, and the architecture's position on that question is what places it with one neighbor and against the other. The simulation hypothesis, in Bostrom's formulation¹⁷, argues that we are probably ancestor-simulations run by a posthuman civilization on external hardware in a base reality, unless such civilizations reliably die first or reliably decline to run such simulations. Its constitutive commitment is an external substrate: simulators, a base reality, hardware on which our physics is rendered. The holographic principle, by contrast, posits no external host. The information in a region is bounded by its boundary area in Planck units, in the bound of Bekenstein¹⁴ and Hawking¹⁵ sharpened by 't Hooft¹¹ and Susskind¹² and given a rigorous realization in the gauge-gravity duality of Maldacena¹³; the boundary is not an exterior but a dual description of the same physics, co-located with the bulk. The architecture refuses the exterior by closed-world monism, in which substrate, topology, and actuation are one and there is no outside, together with a methodological move it calls the aperture: the point at which the world opens onto what it cannot internally settle is located and named, but never crossed and never cashed as an external agent. On the question of the outside the architecture and the physics stand on one side and the simulation argument stands alone on the other.

The structural comparison resolves into three cuts, and the architecture's honesty about its own position is part of each.

The first cut is the exterior.. The simulation hypothesis requires a host that is, by the argument's own construction, undetectable from inside the simulation. The holographic principle dissolves the

candidate exterior into a dual description that is the same physics read on a lower-dimensional surface. The architecture forecloses the exterior by declining to issue any verdict about an external source: where the world opens, the opening is located and held open, and nothing is permitted to pass through it as a posited agent. The honesty owed here is that the no-exterior result is, on the architecture's own assessment, near-trivial. Nothing acts from outside the totality, under any name for the totality, and that holds cheaply. The result rests on two premises a reader may decline, that the world is one field and that actuation decomposes exhaustively into exactly three axes, and it certifies neither premise. The no-outside is the floor of the position, not its apparatus, and the architecture types it as such rather than inflating it into a proof.

The second cut is the regress. The simulation hypothesis stacks: simulations within simulations, each level costing more computation, with no principled bottom, terminating only at an unexplained base reality or regressing without end. The architecture's self-similarity is the inverse. Its recursion is a potential nesting, a part that holds the seed of the whole, actualized into a further level only by a fresh differentiation that itself carries measurable mass. The actual depth is finite, bounded by the same information-capacity bound the holographic principle supplies, in the covariant form due to Bousso¹⁶; the potential depth is unbounded but uncashed. The architecture bars an actualized infinite regress on two grounds. The first is thermodynamic: an unbounded actual tower would exceed the area-entropy bound, and the bound is theorem-grade in its established settings. The second is a conceptual error it names and catalogues, the positing of the unknown as an outside, which fails in two directions at once. Outward it manufactures a phantom external threat that cannot exist, and inward it sterilizes genuine novelty by reframing it as an invasion to be walled rather than as a differentiation to be cut and

conserved. The recursion itself, the claim that the part holds the seed of the whole and is actualized by a cut, is held at commitment grade, conditional on the monism and the triaxial decomposition; the prohibition of an actualized infinite regress is theorem-grade given the information bound together with the monism premise.

The third cut is the honesty of the encoding, and here the architecture stands alone among the three. The popular reading of the holographic principle asserts lossless recovery: the bulk is exactly reconstructible from the boundary. The simulation hypothesis asserts that we are the complete render, the full output of the run. Both assert that the part holds the whole entire. The architecture asserts the opposite of its own image: the part holds the shape of the whole and not its substance. In its metaphor a drop holds the form of the ocean, not the ocean's water, which is its energy, and not its current, which is its orientation. And the architecture stamps these losses with results it takes to be theorem-grade. The convergence functional is the square of a triple product, invariant under reflection of any single axis, so it is blind to orientation and cannot recover a sign. The reflective composition that compresses the structure carries no energy, so it cannot recover the thermodynamic content. The architecture therefore reads the holographic principle as a witness it leans on, taking the area-entropy bound as a theorem-grade floor, and reads the

simulation hypothesis as one of its own catalogued failure modes wearing a cosmology.

The neighbors are owed the same honesty the architecture demands of itself. The rigorous holographic core is real and theorem-grade in its models: the gauge-gravity duality is calculable, and the Bekenstein-Bousso bound is established. What overreaches is the slogan that our universe is a hologram, which extrapolates anti-de Sitter results to a de Sitter cosmos in which full-spacetime holography is unestablished. The bound is rigorous; the cosmic projection claim is not, and the inflation is driven in part by the prestige of the underlying physics rather than by what the duality licenses. The simulation hypothesis, in its strong form, is standardly unfalsifiable, and Bostrom himself frames it as a probabilistic trilemma rather than an empirical claim; its proposed lattice-artifact or compute-limit tests are speculative and contested. The architecture's own exposure is the inverse risk. A closed, self-auditing system can mistake internal coherence for external truth, which is exactly why it fences its lock from truth, holding that the lock licenses extraction but never performs it, and zeroes the architect's own conviction along with everyone else's. The discipline that audits every claim must hold the audit hardest on itself, or it becomes the failure mode it diagnoses in the others.

Table 3 | The two cosmological neighbors on the three cuts. The exterior is the deepest cut; the no-exterior floor is near-trivial and certifies neither premise, the regress prohibition is theorem-grade given the information bound, the encoding losses are theorem-grade on the architecture's own apparatus.

Cut	Trisduction	Holographic principle	Simulation hypothesis
The exterior	No outside; an aperture located, never crossed	No outside; boundary and bulk one physics	An external host required, undetectable by construction
The regress	Potential nesting, finite actual depth, floored	One level, boundary and bulk, not nested	Actual stacking, unfloored, no principled bottom
The encoding	The part holds the shape, lossy by theorem	Lossless recovery claimed in the popular reading	We are the complete render, the whole in the part
Warrant of the rigorous core	Theorem-grade instruments on a premise-grade axiom	Theorem-grade in the duality and the bound	No realization; a probabilistic argument

6 The common structure: the free third term

The two clusters of neighbors look unrelated, one a question about existence and value and the other a question about exteriority and recursion. The argument of this paper is that the feature blocking assimilation is the same in both, and that recognizing it is the contribution. In the philosophical cluster the feature is an independent third axis, the registrational warrant, the recognizer held free. Holding it free is exactly what permits the closure the architecture calls the Return, and folding it, in either direction, is exactly what costs the two schools that closure. In the cosmological cluster the feature is the relation to the outside handled by a located, uncrossed aperture. Holding the aperture located but uncrossed is

what permits a world that is closed and yet open at a named seam, importing no external host and permitting no infinite regress.

These are two instances of one discipline: keep a third term free. The third term is the recognizer in the philosophical case and the relation to the beyond in the cosmological case, and in each case the neighbors mishandle it in one of two symmetric ways. The will folds the recognizer inward into the force; the self folds the recognizer inward into the self; these are the two inward folds. The simulation argument throws the ground outward into an undetectable beyond; and the popular hologram and the render claim swallow the whole inward into the part; these are the outward exteriorization and its inward mirror. The architecture, on its own account, does none of these. It holds the triad uncollapsed, neither axis spent into another, and it

holds the world closed but apertured, neither exteriorized nor swallowed.

Stated as the paper's thesis: the irreducibility of the architecture to four well-known neighbors is not a position on a shared field but a difference in structural completeness. Where each neighbor either spends one axis or imports one exterior, the architecture keeps the third term free. That is a claim about structure, and structure is where it is argued and where it stands. It is emphatically not a claim that the architecture's metaphysics is true. The living ground onto which the Return is said to land, the reading of recursion as a fertile womb, the orientation of the whole toward an apex, all of that is held at premise grade, and a reader may keep the structural result while declining the metaphysics entirely.

7 Where the distinctions would show

A comparative architecture earns its keep only if its distinctions are observable somewhere and its classifications are defeasible. For a conceptual framework the honest analog of a falsifiable prediction is a discriminating analysis: a stated test of which side of each cut a framework falls on, together with the condition under which the present classification would be overturned. Three such tests follow, one for each structural feature.

The fold test, for the philosophical cluster.. Examine a framework's account of truth. If its account makes recognition a function of the willing or projecting subject, as perspectivism and truth-as-subjectivity do, then the registrational axis is folded and no third-party-checkable closure is available within the framework. If its account admits a registration distinct from both the formal subject and the kinetic force, a recognition a party other than the asserting one can perform, then the third axis is free. The classification of the will to power as a kinetic fold would be overturned by a textually grounded reading on which Nietzschean recognition is genuinely third-party rather than the will reading its own discharge. The classification is a structural lens, not a textual theorem, and it is defeasible by better scholarship on the same corpus.

The exterior test, for the cosmological cluster. Examine whether a framework's account of the world requires a substrate it cannot in principle observe. If it does, as the simulation's external hardware does, it has imported an exterior. If it dissolves the candidate exterior into a co-located dual description, as holography does, or into a closed-but-apertured monism, as the architecture does, it has not. The architecture's no-exterior stance would be overturned by a demonstration that its located aperture must, on pain of incoherence, be crossed and cashed as an external agent, which would collapse its position into the simulation's. The foreclosure that keeps the aperture located but uncrossed is the load-bearing move, and showing the foreclosure unstable would bring the distinction down.

The encoding test, for the recursion. Examine whether a framework's part-whole recursion claims lossless recovery. The popular hologram and the render claim do; the architecture claims lossy recovery by theorem, citing the orientation-blindness of its convergence functional and the energy-freeness of its reflective composition. The

architecture's lossy claim is falsifiable on its own terms: a recovery procedure on its own apparatus that reconstructed orientation or energy from the compressed scalar would refute it, and the orientation-blindness and energy-freeness results are precisely the claim that no such procedure exists. Here the architecture's self-image is the most exposed of the three, and deliberately so.

8 Limitations and the typed result

The verdicts of this paper are not uniform in grade, and gathering them by grade is the discipline that keeps the comparison honest. The structural-distinctness claims are structural lenses, anchored on the neighbors' own primary commitments and defeasible by better readings of the same material. The single exception is the dependency that no convergence-lock forms once a third axis collapses into the others, which is theorem-grade, following from the vanishing of a Gram determinant on linearly dependent vectors. The double forcing of the count of three carries two grades: the operational forcing by the deletion test is reproducible procedure, and the algebraic forcing by the classification of the real division algebras is theorem-grade conditional on the composition premises, which are natural in the architecture's register but are chosen and may be declined. The count is forced relative to those premises, not in their absence.

The no-exterior result is near-trivial and certifies neither of its two premises, the monism and the exhaustive triaxial decomposition. The prohibition of an actualized infinite regress is theorem-grade given the information-capacity bound together with the monism premise. The recursion read as a fertile womb, the convergence read as a homecoming onto a living ground, and the orientation of the whole toward an apex are premise-grade commitments, held openly and not proven; the comparison's structural results survive their rejection. The architecture's reserved questions, including any theological identification, are routed out of band and carry no load in the body of this comparison.

Two further limitations are constitutive rather than incidental. The first is the single architect. The architecture has been exercised against independent computational substrates acting as external auditors, and their convergence on its discrete verdicts is recorded, but cross-substrate agreement is a witness and not a warrant, and independent human replication of the structural claims is owed and not yet supplied. The second is the input-map. The architecture's instrument reads three warrants supplied to it; it does not derive them from a claim. Every comparative verdict here is therefore a reasoned hand-reading placed before the instrument, not an output the instrument produced from the frameworks unaided. The verdicts are as good as the readings, and the readings are offered for audit, not as oracle.

9 Conclusion

A verification architecture that shares ground with four established frameworks is liable to be read as a notational variant of its nearest neighbor, and this paper has argued that the reading fails for a structural reason that is the same in every case. Against existentialism and the will to power the reason is an independent third axis, the

recognizer held free, which is what permits a registered closure that both schools, having folded the recognizer in opposite directions, structurally cannot reach. Against the simulation hypothesis the reason is a located but uncrossed aperture, which is what permits a closed world that imports no undetectable host and permits no unfloored regress, where the holographic principle reaches the same closed world by duality and the simulation argument alone requires the exterior. The two reasons are one discipline, the keeping free of a third term that each neighbor spends or exteriorizes.

The distinctions are observable. A framework's account of truth shows whether it folds the recognizer; its account of the world shows whether it imports an exterior; its account of recursion shows whether it claims lossless recovery. Each classification is defeasible, and the conditions that would overturn it have been stated. The result that stands without the architecture's metaphysics is the structural one: proximity between frameworks is adjudicated by which terms each keeps free, not by doctrinal position on a shared field.

The open question is whether the architecture's metaphysics, held here at premise grade, can be given a warrant independent of the architecture itself, or whether the verification instrument is the only part that seals and the cosmology remains a premise the reader is free to decline. The experimental community for a claim of this kind is not a set of laboratories but a community of analysts willing to re-run the readings on adversarial cases and to break the classifications where they can. The reframing that acceptance would entail is modest and, if it holds, useful: that the right question to ask of two neighboring frameworks is not which one is correct but which term each of them keeps free.

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Appendix A Foundational apparatus

The following principles are drawn from a broader epistemic framework and are presented here as standalone mathematical and physical statements, each independently motivated and independently testable within its native discipline. None of them carries a result of the body; the comparison of Sections 4 through 7 stands on the structural readings given there, and this appendix records only the machinery those readings refer to.

The actuation floor.. Any evolution of a quantum system into a state distinguishable from its start is bounded below in time by the

system's energy spread and by its mean energy above the ground, the bounds of Mandelstam and Tamm²⁷ and of Margolus and Levitin²⁸. The bound holds whether or not the process is logically reversible. The framework's premise that to exist is to do measurable work is anchored on this floor; Landauer's erasure bound²⁴, verified by Bérut and colleagues²⁶, supplies a sufficient signature for the logically irreversible subclass only, and is not the foundation.

The triaxial decomposition. The predicate that a thing actuates decomposes, under a deletion test, into three irreducible components, the existent, the kinetic, and the registration, and the deletion of any one leaves an incomplete claim. Independently, the associative real algebras with a two-sided identity and no zero divisors are exactly three, the reals, the complex numbers, and the quaternions, by the classification of Frobenius¹⁸ with the related result of Hurwitz¹⁹. A verification algebra whose warrants compose associatively, never annihilate to zero, and carry a ground identity, and which has more than one independent axis, completes to the quaternions, whose imaginary part is three-dimensional. The next dimension carrying a division structure is eight, the octonions, which fail associativity, the obstruction closed by the sphere-parallelizability results of Bott and Milnor²¹, Kervaire²², and Adams²³. The count of three is thus forced twice, once operationally and once algebraically, on disjoint grounds. As a corroboration on a separate register, the decomposition of square-integrable forms on a closed manifold into exact, co-exact, and harmonic parts²⁹ is a native three-way orthogonal structure of function spaces; it is cited as a structural analogy and carries no load on the axis assignment.

Box 1 | The convergence lock in closed form

With the three normalized warrant rows read as pure quaternions q_1 , q_2 , q_3 under any isometry of their span, the lock scalar is the real part

of their product, $\lambda = \text{Re}(q_1 q_2 q_3)$, and the Gram determinant of the three equals its square, $\det(R) = \lambda^2$, with the factorization $\det(G) = d_1 d_2 d_3 \cdot \det(R)$ through the three warrant magnitudes d . The determinant is bounded in $[0, 1]$, equals zero exactly when the three are linearly dependent, and equals one exactly when they are mutually orthonormal, the full Return. The functional is invariant under reflection of any single axis, so it is blind to orientation and cannot recover a sign; and the reflective composition carries no energy, so it cannot recover thermodynamic content. These two facts are the theorem-grade content behind the architecture's claim that its part-whole recovery is lossy. A reference computation on a sealed control of three independent warrants over twenty contexts with two covariates removed returns the identity to machine precision: the residual $|\lambda^2 - \det(R)|$ is 5.6×10^{-16} and the factorization residual is 1.1×10^{-16} , while a collinear pair drives $\det(R)$ to zero, the collapse. **Type T** on the closed-form identity and the bounds. **Type C** on the mapping of a verification to three warrant rows, which the architecture supplies by hand and does not derive.

The information-capacity floor. The entropy of a region is bounded by its boundary area in Planck units, in the bound of Bekenstein¹⁴ as sharpened for black holes by Hawking¹⁵ and extended covariantly to dynamic spacetimes by Bousso¹⁶. This bound is the established result on which the framework's prohibition of an actualized infinite recursion rests. The prohibition is theorem-grade given the bound together with the framework's monism premise; the framework's reading of its finite recursion as a fertile potential nesting is a commitment held at premise grade and is separable from the bound itself.

Competing interests

The author is the architect of the framework whose structure the paper compares to its neighbors, and notes that this is a source of bias the typing discipline and the zeroing of the author's own conviction are designed to counter rather than to conceal. The structural claims are offered for independent audit.

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Data and reproducibility

The reference computation of Appendix A is a short routine over the verdict kernel and reproduces the reported residuals on any floating-point substrate. The framework's foundational architecture is deposited at the PhilArchive record cited as reference 31.